

IoT Dashboard Project

DHT11 + LDR + Node-RED Dashboard

Complete Node-RED Setup & Documentation

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Technology	Tool / Platform
Microcontroller	ESP12-E (ESP8266)
Sensors	DHT11 (Temp & Humidity), LDR (Light)
Protocol	MQTT (Message Queuing Telemetry Transport)
MQTT Broker	HiveMQ (broker.hivemq.com)
Flow Platform	Node-RED
Dashboard	Node-RED Dashboard (/ui)
Programming IDE	Arduino IDE

1. Project Overview

This document provides a complete technical explanation of the IoT Dashboard project built using ESP8266 (ESP12-E), DHT11 temperature & humidity sensor, LDR light sensor, MQTT protocol with HiveMQ broker, and Node-RED dashboard for real-time visualization.

1.1 Project Goals

- Read real-time Temperature, Humidity, and Light data from sensors
- Transmit sensor data wirelessly via WiFi using MQTT protocol
- Receive and process data in Node-RED flow editor
- Display live data on a professional Node-RED Dashboard with Gauges and Charts
- Store and visualize historical data trends using line charts

1.2 System Architecture

The complete system follows this data flow:

Layer	Component	Role
Hardware	DHT11 + LDR Sensors	Measure temperature, humidity, light
Controller	ESP12-E (ESP8266)	Read sensors, connect WiFi, publish MQTT
Network	WiFi Router	Internet connectivity
Broker	HiveMQ (broker.hivemq.com)	Receive and forward MQTT messages
Flow Engine	Node-RED	Subscribe to MQTT, process data
Visualization	Node-RED Dashboard /ui	Show Gauges and Live Charts

2. Hardware Setup

2.1 Components Required

Component	Specification	Quantity
ESP12-E (ESP8266)	WiFi SoC, 80MHz, 1MB Flash	1
DHT11 Sensor	Temp: 0-50°C, Humidity: 20-90%	1
LDR Sensor	Light Dependent Resistor, 5mm	1
10K Resistor	Voltage divider for LDR	1
USB-UART Converter	CH340 or FTDI FT232	1
Breadboard	830 tie-point	1
Jumper Wires	Male-to-Male	As needed
3.3V Power Supply	Via USB-UART or regulator	1

2.2 Pin Connections

Sensor	Sensor Pin	ESP12-E Pin	Description
DHT11	VCC	3.3V	Power supply
DHT11	GND	GND	Ground
DHT11	DATA	D2 (GPIO4)	Data signal pin
LDR	Leg 1	3.3V	Power supply
LDR	Leg 2	A0 (ADC)	Analog input
10K Resistor	One end	LDR Leg 2	Pull-down resistor
10K Resistor	Other end	GND	To ground

3. MQTT Protocol

3.1 What is MQTT?

MQTT (Message Queuing Telemetry Transport) is a lightweight, publish-subscribe messaging protocol designed for IoT devices. It is ideal for low-bandwidth, high-latency, or unreliable networks.

3.2 MQTT Key Concepts

Concept	Description
Broker	Central server that receives and distributes all messages (HiveMQ)
Publisher	Device that sends data to a topic (ESP8266)
Subscriber	Device that receives data from a topic (Node-RED)
Topic	Channel name for messages (e.g., tempNode, humiNode)
QoS 0	Fire and forget — message sent once, no guarantee
QoS 1	At least once — message delivery guaranteed
QoS 2	Exactly once — highest guarantee, slowest

3.3 MQTT Topics Used in This Project

Topic Name	Sensor	Data Type	Example Value	Unit
tempNode	DHT11	Float	28.50	°C
humiNode	DHT11	Float	65.00	%
ldrNode	LDR	Integer	512	lux (0-1024)

3.4 HiveMQ Broker Configuration

Parameter	Value
Broker Host	broker.hivemq.com
Port	1883 (TCP) / 8883 (SSL)
Authentication	None (Public broker)
Protocol	MQTT V3.1.1
Keep Alive	60 seconds
Client ID	ESP8266-[random hex] (auto-generated)

4. Node-RED — Complete Guide

4.1 What is Node-RED?

Node-RED is a flow-based, visual programming tool built on Node.js. It allows you to connect hardware devices, APIs, and online services using a browser-based editor. Each function is represented as a 'node', and nodes are connected with 'wires' to create a flow.

4.2 Node-RED Interface

UI Element	Location	Purpose
Node Palette	Left Panel	Contains all available nodes to drag onto canvas
Canvas	Center Area	Work area where nodes are placed and connected
Config Panel	Right Panel	Shows node properties, debug output, settings
Deploy Button	Top Right	Activates/saves the current flow
Debug Tab	Right Panel	Shows real-time output from debug nodes
Dashboard Tab	Right Panel	Shows layout of UI dashboard widgets

4.3 Node Types Used

Node	Category	Color	Purpose
mqtt in	Network	Purple	Subscribes to MQTT topic and receives data
debug	Common	Green	Prints message payload in debug panel
gauge	Dashboard	Teal	Displays a circular gauge widget on dashboard
chart	Dashboard	Teal	Displays a line/bar chart widget on dashboard
function	Function	Yellow	Runs custom JavaScript to process data

4.4 How Node-RED Flow Works

The flow in this project follows this pattern:

```
mqtt in (tempNode) → gauge (Temperature)
                    → chart (Temperature Graph)

mqtt in (humiNode) → gauge (Humidity)
```

```

    → chart (Humidity Graph)

mqtt in (ldrNode) → gauge (Light)
                  → chart (Light Graph)

```

4.5 MQTT In Node Configuration

The mqtt in node subscribes to a specific topic on the MQTT broker. When a message is published to that topic, this node receives it and passes it to the next node via a wire.

Field	Value	Description
Server	HiveMQ	MQTT broker configuration
Action	Subscribe to single topic	Listen to one topic
Topic	tempNode / humiNode / ldrNode	Topic to subscribe
QoS	2	Message delivery guarantee level
Output	Auto-detect (JSON/string/buffer)	Auto parse incoming data
Name	Temperature / Humidity / LDR Sensor	Node label on canvas

4.6 Gauge Node Configuration

The gauge node displays a semicircular meter on the dashboard. It shows the current sensor value in real-time.

Field	Temperature	Humidity	Light
Group	[IoT Dashboard] Sensors	[IoT Dashboard] Sensors	[IoT Dashboard] Sensors
Type	Gauge	Gauge	Gauge
Label	Temperature	humidity	light
Unit	°C	%	lux
Min Range	0	0	0
Max Range	50	100	1024
Size	4 x 4	4 x 4	4 x 4

4.7 Chart Node Configuration

The chart node displays a live line graph of historical sensor values with timestamps on the X-axis.

Field	Temperature	Humidity	Light
Group	[Charts] chart	[Charts] chart	[Charts] chart
Type	Line Chart	Line Chart	Line Chart
Label	Temperature Graph	Humidity Graph	Light Graph
X-axis	Last 1 hour	Last 1 hour	Last 1 hour
Y-axis	Auto	Auto	Auto
Size	4 x 4	4 x 4	4 x 4

4.8 Dashboard Pages

Page	URL	Content	Widgets
IoT Dashboard	127.0.0.1:1880/ui/#/0	Live sensor gauges	3 Gauge widgets side by side
Chart	127.0.0.1:1880/ui/#/1	Historical graphs	3 Line chart widgets side by side

5. Installation & Setup Guide

5.1 Node-RED Installation

Step 1: Install Node.js from nodejs.org

Step 2: Open Command Prompt and run:

```
| npm install -g --unsafe-perm node-red
```

Step 3: Start Node-RED:

```
| node-red
```

Step 4: Open browser and go to:

```
| http://127.0.0.1:1880
```

5.2 Node-RED Dashboard Installation

In Node-RED editor:

- Click Menu (top right) → Manage Palette
- Click Install tab
- Search: node-red-dashboard
- Click Install button
- Wait for installation to complete
- Restart Node-RED if prompted

5.3 Arduino IDE Setup for ESP8266

Step 1: Install ESP8266 Board:

- Open Arduino IDE → File → Preferences
- Add URL: http://arduino.esp8266.com/stable/package_esp8266com_index.json
- Tools → Board → Board Manager → Search ESP8266 → Install

Step 2: Install Libraries:

- Sketch → Include Library → Manage Libraries
- Search and install: DHT sensor library (Adafruit)
- Search and install: PubSubClient (Nick O'Leary)

Step 3: Board Settings:

Setting	Value
Board	Generic ESP8266 Module
Upload Speed	115200
CPU Frequency	80 MHz
Flash Size	1MB (FS: 64KB OTA: ~470KB)
Port	COM port of USB-UART converter

6. Troubleshooting Guide

Problem	Possible Cause	Solution
WiFi not connecting	Wrong SSID/Password	Check credentials in code
MQTT not connecting	Broker unreachable	Check internet connection, try different broker
DHT sensor error	Wrong pin or bad connection	Check D2 pin wiring, add 10K pull-up
LDR always 0	No voltage divider	Add 10K resistor between LDR and GND
Dashboard not loading	node-red-dashboard not installed	Install via Manage Palette
Gauge shows 0	Topic mismatch	Check MQTT topic names match exactly
No data in debug	Flow not deployed	Click Deploy button

7. Project Summary

7.1 What Was Achieved

- Successfully built a complete WiFi-based IoT monitoring system
- ESP8266 reads DHT11 and LDR sensors every 5 seconds
- Data published to HiveMQ public MQTT broker
- Node-RED subscribes and processes incoming sensor data
- Live dashboard with 3 gauges and 3 line charts
- Two separate dashboard pages — Sensors and Charts

7.2 Key Learning Outcomes

Topic	What Was Learned
MQTT Protocol	Publish-Subscribe model, topics, QoS levels, broker
HiveMQ	Public MQTT broker configuration and connection
Node-RED	Flow editor, node types, wires, canvas, deploy
Dashboard	Gauge nodes, chart nodes, groups, tabs, layout
ESP8266	WiFi connection, MQTT library, sensor reading
IoT Architecture	End-to-end data flow from sensor to dashboard

7.3 Author Details

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